

Calculators, cell phones, or notes are not allowed.

- (1) The continued fraction algorithm for $\alpha > 1$ is purely periodic of period 2 such that $[\alpha] = [\alpha_2] = [\alpha_4] = 6$ and $[\alpha_1] = [\alpha_3] = [\alpha_5] = 3$, thus

$$\alpha = 6 + \frac{1}{3 + \frac{1}{6 + \dots}}$$

Find α .

We have:

$$\alpha = 6 + \frac{1}{3 + \frac{1}{\alpha}} = 6 + \frac{\alpha}{3\alpha + 1} = \frac{19\alpha + 6}{3\alpha + 1}.$$

Thus $3\alpha^2 - 18\alpha = 6$ i.e. $(\alpha - 3)^2 = 11$ i.e. $\alpha = 3 + \sqrt{11}$ (as $\alpha > 6$).

- (2) Use the previous exercise to find all solutions to $X^2 - 11Y^2 = 1$.

We have

$$3 + \sqrt{11} = 6 + \frac{1}{3 + \frac{1}{3 + \sqrt{11}}} \quad \text{or} \quad \sqrt{11} = \frac{10\sqrt{11} + 33}{3\sqrt{11} + 10}.$$

Thus $(10, 3)$ is the first solution and all other solutions are given by $(10 + 3\sqrt{11})^k$, with $k = 1, 2, 3, \dots$